1. Load the dataset and convert to DataFrame

```
#Mohanasundaram
#241801169
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer
from sklearn.model_selection import train_test_split # Fixed the import module to model_selection

# Let's download the titanic dataset first if not already present
url = 'https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv'
df = pd.read_csv(url)
display(df.head())
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

2. Understand the data (shape, info, describe)

```
print("Shape of dataset:", df.shape)
print("\n--- Dataset Info ---")
df.info()
print("\n--- Descriptive Statistics ---")
display(df.describe(include='all'))
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/3057031436.py in <cell line: 0>()
----> 1 print("Shape of dataset:", df.shape)
      2 print("\n--- Dataset Info ---")
      3 df.info()
      4 print("\n--- Descriptive Statistics ---")
      5 display(df.describe(include='all'))

NameError: name 'df' is not defined
```

3. Fill missing values in Age using SimpleImputer

4. Fill missing Cabin with "Unknown"

5. Fill missing Embarked with the mode

```
# 3. Age Imputation
imputer = SimpleImputer(strategy='median')  # Median is robust to outliers for Age
df['Age'] = imputer.fit_transform(df[['Age']])

# 4. Fill Cabin with 'Unknown'
df['Cabin'] = df['Cabin'].fillna('Unknown')

# 5. Fill Embarked with mode
embarked_mode = df['Embarked'].mode()[0]
df['Embarked'] = df['Embarked'].fillna(embarked_mode)

# Verify no missing values in these columns anymore
print(df[['Age', 'Cabin', 'Embarked']].isna().sum())
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/1765347942.py in <cell line: 0>()
      1 # 3. Age Imputation
      2 imputer = SimpleImputer(strategy='median')  # Median is robust to outliers for Age
----> 3 df['Age'] = imputer.fit_transform(df[['Age']])
      4
      5 # 4. Fill Cabin with 'Unknown'

NameError: name 'df' is not defined
```

6. Analyse the passenger Class feature using countplot

```
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Pclass', palette='Set2')
plt.title('Passenger Distribution by Class (Pclass)')
plt.xlabel('Passenger Class')
plt.ylabel('Count')
plt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/2020449807.py in <cell line: 0>()
      1 plt.figure(figsize=(8, 5))
----> 2 sns.countplot(data=df, x='Pclass', palette='Set2')
      3 plt.title('Passenger Distribution by Class (Pclass)')
      4 plt.xlabel('Passenger Class')
      5 plt.ylabel('Count')

NameError: name 'df' is not defined

<Figure size 800x500 with 0 Axes>
```

7. List the names of passengers who are female and survived

```
female_survived = df[(df['Sex'] == 'female') & (df['Survived'] == 1)]['Name']
display(female_survived.to_frame())
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/2508831409.py in <cell line: 0>()
----> 1 female_survived = df[(df['Sex'] == 'female') & (df['Survived'] == 1)]['Name']
      2 display(female_survived.to_frame())

NameError: name 'df' is not defined
```

8. Find all passengers who are in 3rd class and under the age of 18

```
class3_under18 = df[(df['Pclass'] == 3) & (df['Age'] < 18)]
display(class3_under18)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/2097387974.py in <cell line: 0>()
----> 1 class3_under18 = df[(df['Pclass'] == 3) & (df['Age'] < 18)]
      2 display(class3_under18)

NameError: name 'df' is not defined
```

9. Find the list of passengers who are in class 1 and older than 40

10. Find the list of passengers who survived in above category

```
# 9. Class 1 and older than 40
class1_over40 = df[(df['Pclass'] == 1) & (df['Age'] > 40)]
print(f"Number of passengers in class 1 and > 40 years old: {len(class1_over40)}")
display(class1_over40.head())
```

```
# 10. Survived in above category
class1_over40_survived = class1_over40[class1_over40['Survived'] == 1]
print(f"\nNumber of survivors among them: {len(class1_over40_survived)}")
display(class1_over40_survived.head())
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/1033001185.py in <cell line: 0>()
      1 # 9. Class 1 and older than 40
----> 2 class1_over40 = df[(df['Pclass'] == 1) & (df['Age'] > 40)]
      3 print(f"Number of passengers in class 1 and > 40 years old: {len(class1_over40)}")
      4 display(class1_over40.head())
      5

NameError: name 'df' is not defined
```

11. Which male passengers paid a fare greater than 100?

12. List passengers who embarked from port ‘C’ and were in Class 2

```
# 11. Male, fare > 100
male_high_fare = df[(df['Sex'] == 'male') & (df['Fare'] > 100)]
print("Male passengers paying fare > 100:")
display(male_high_fare)

# 12. Embarked port 'C' and Class 2
port_c_class2 = df[(df['Embarked'] == 'C') & (df['Pclass'] == 2)]
print("\nPassengers from Port C in Class 2:")
display(port_c_class2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/2116519836.py in <cell line: 0>()
      1 # 11. Male, fare > 100
----> 2 male_high_fare = df[(df['Sex'] == 'male') & (df['Fare'] > 100)]
      3 print("Male passengers paying fare > 100:")
      4 display(male_high_fare)
      5

NameError: name 'df' is not defined
```

13. Show all passengers who had more than 2 siblings/spouses aboard (SibSp > 2)

14. Passengers who did not survive and had no family aboard (SibSp = 0 and Parch = 0)

```
# 13. SibSp > 2
sibsp_gt2 = df[df['SibSp'] > 2]
print("Passengers with more than 2 siblings/spouses:")
display(sibsp_gt2)

# 14. Did not survive & no family
no_survive_no_family = df[(df['Survived'] == 0) & (df['SibSp'] == 0) & (df['Parch'] == 0)]
print(f"\nNumber of passengers who did not survive and had no family: {len(no_survive_no_family)}")
display(no_survive_no_family.head())
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/1535698044.py in <cell line: 0>()
      1 # 13. SibSp > 2
----> 2 sibsp_gt2 = df[df['SibSp'] > 2]
      3 print("Passengers with more than 2 siblings/spouses:")
      4 display(sibsp_gt2)
      5

NameError: name 'df' is not defined
```

```
# 15. Top 5 oldest survivors
top5_oldest_survivors = df[df['Survived'] == 1].sort_values(by='Age', ascending=False).head(5)
print("Top 5 oldest survivors:")
display(top5_oldest_survivors)

# 16. Zero fare passengers
zero_fare = df[df['Fare'] == 0]
print(f"\nNumber of passengers who paid zero fare: {len(zero_fare)}")
display(zero_fare)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/324164420.py in <cell line: 0>()
      1 # 15. Top 5 oldest survivors
----> 2 top5_oldest_survivors = df[df['Survived'] == 1].sort_values(by='Age', ascending=False).head(5)
      3 print("Top 5 oldest survivors:")
      4 display(top5_oldest_survivors)
      5

NameError: name 'df' is not defined
```

17. Split the dataset for training and testing

```
from sklearn.model_selection import train_test_split

# Define features (X) and target variable (y)
X = df.drop(columns=['Survived'])
y = df['Survived']

# Split into train (80%) and test (20%) sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print(f"Training set shape (X_train): {X_train.shape}")
print(f"Testing set shape (X_test): {X_test.shape}")
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1248/2510327477.py in <cell line: 0>()
      2
      3 # Define features (X) and target variable (y)
----> 4 X = df.drop(columns=['Survived'])
      5 y = df['Survived']
      6

NameError: name 'df' is not defined
```